KCE PATHWAYS · GRADES 4–5 · UNIT 6: VOICES OF CHANGE · WEEK 28 · MATH

Race Day in Boulder

Name: _____ **Date:** _____

PRE-LESSON WARM-UP · DO THIS BEFORE YOUR COMPUTER LESSON

Pre-Lesson Warm-Up: Decimals on the Course

Lavinia runs cross-country in Boulder, Colorado, where race times and distances are full of decimals. Warm up before the lesson, and show your work. Some problems may feel new, and that is okay!

1. Review (Week 27): a) 4.6×10^2; b) $53 \div 10^3$; c) Write $10 \times 10 \times 10 \times 10 \times 10$ with an exponent.

2. Multiply the fractions: 3/10 × 7/10. Then write your answer as a decimal.

3. Estimate by rounding each number to the nearest whole number: a) 3.9 × 2.1; b) 6.2 × 4.8

4. How many 0.5-kilometer sections fit in a 3-kilometer course?

   6 | 1.5 | 0.6

5. Leandro says 0.4 × 0.2 = 0.8. Do you agree? Explain your thinking.

KCE PATHWAYS · GRADES 4–5 · UNIT 6: VOICES OF CHANGE · WEEK 28 · MATH

Race Day in Boulder

Name: _____ **Date:** _____

POST-LESSON CHALLENGE · DO THIS AFTER YOUR COMPUTER LESSON

Post-Lesson Challenge: The Coach's Clipboard

Estimate first, count the decimal places, and make the divisor a whole number before you divide. Show your work.

1. Multiply. a) 0.7 × 0.6; b) 2.3 × 1.4; c) 4.5 × 0.8

2. Divide. a) 5.6 ÷ 0.7; b) 0.81 ÷ 0.09; c) 7.5 ÷ 1.5

3. Two steps: Lanelle runs 3 kilometers in 16.2 minutes. What is her pace in minutes per kilometer? At that pace, how many minutes would 4.5 kilometers take?

4. Find the mistake: Leandro says 3.2 ÷ 0.8 = 0.4. In two or three complete sentences, explain his mistake and give the correct quotient.

★ **Challenge:** Without computing, put these in order from least to greatest: 7.2 × 0.9, 7.2 ÷ 0.9, and 7.2. Explain how you know. Then compute the product and the quotient to check.

Looking Ahead: Week 29: Multiplying Fractions as Scaling. With tiny-house builders in Austin, Texas, you will predict whether a product is bigger or smaller and find the area of rectangles with fractional side lengths.

KCE PATHWAYS · GRADES 4–5 · UNIT 6: VOICES OF CHANGE · WEEK 28 · READING

Wings Over Michoacán

Name: _____ **Date:** _____

PRE-LESSON WARM-UP · DO THIS BEFORE YOUR COMPUTER LESSON

Pre-Lesson Warm-Up: Thinking Like a Researcher

Today Naveen and Nayeli will read three sources to answer one question about the monarch butterflies that fly to Michoacán, Mexico. Show what you already know. It is fine to guess!

1. Review (Week 27): Nanami's pencil case is missing. Her cousin Nasir has a pencil case just like it, but his has his name inside. Later she finds hers in the car. Which detail is the red herring, and why?

2. What do you already know about monarch butterflies? Write two facts, and tell where you learned each one.

3. Which of these is a **research question**?

   Monarchs are orange. | Where do monarchs spend the winter? | I saw a monarch today.

4. Why might a researcher read more than one source before writing an answer?

5. Read: "Source 1 says the pond freezes in December. Source 2 says ducks leave the pond when it freezes." Write ONE sentence that uses both facts and names both sources.

KCE PATHWAYS · GRADES 4–5 · UNIT 6: VOICES OF CHANGE · WEEK 28 · READING

Wings Over Michoacán

Name: _____ **Date:** _____

POST-LESSON CHALLENGE · DO THIS AFTER YOUR COMPUTER LESSON

Post-Lesson Activity: Research Notes

Write in complete sentences with correct capitals, commas, and quotation marks. Name your sources.

1. Two steps. Which source in the lesson gave the acres the colonies covered? Use it to explain whether the monarchs have fully recovered.

2. Find the mistake. Naveen wrote: "Monarchs lay their eggs on oyamel fir trees in Mexico (Source A)." Explain what is wrong and fix the sentence.

3. One source says a fact is "about eight months," and another says "8 to 9 months." Explain what a careful researcher should do, and why.

4. Be the editor. Rewrite correctly. a) According to Source B the super generation lives longer then its parents. b) There were less butterflies.

★ **Challenge:** Write 2 or 3 sentences: Why do eastern monarchs fly to Michoacán, and how are they doing today? Cite at least two sources.

Looking Ahead: Week 29: Flagstaff, Arizona, debates a new skate park. You will weigh both sides and write a persuasive letter to the city council.

KCE PATHWAYS · GRADES 4–5 · UNIT 6: VOICES OF CHANGE · WEEK 28 · SCIENCE

Heat and Thermal Energy

Name: _____ **Date:** _____

PRE-LESSON WARM-UP · DO THIS BEFORE YOUR COMPUTER LESSON

Pre-Lesson Warm-Up: What Do I Already Know?

Answer with what you know right now. It is OK to guess!

1. **Review (Week 27):** About 2.5 percent of Earth's water is fresh. Where is most of that fresh water stored? Name the two biggest places.

2. Predict: You hold an ice cube and your hand gets cold. What do you think is moving, and in which direction?

3. A metal spoon and a wooden spoon sit in the same bowl of hot soup. Which handle do you think gets hot first? Why?

4. Name two ways your family keeps food or drinks hot or cold. What do you think makes them work?

5. Draw how you think the Sun's warmth reaches Earth across space. Label your drawing.

KCE PATHWAYS · GRADES 4–5 · UNIT 6: VOICES OF CHANGE · WEEK 28 · SCIENCE

Heat and Thermal Energy

Name: _____ **Date:** _____

POST-LESSON CHALLENGE · DO THIS AFTER YOUR COMPUTER LESSON

Post-Lesson Activity: Following the Heat

Use evidence from today's lesson. Write in complete sentences.

1. **Sort it.** Name the way heat moves in each: (a) a pot handle heating on the stove, (b) warm air rising from a heater vent, (c) your face warming in the sunshine.

2. **Two-step.** Change 15 degrees Celsius into Fahrenheit (multiply by 1.8, then add 32). Show both steps.

3. **Find the mistake.** Pieter says, "My mittens make heat for my hands." Explain what is wrong, using the word *insulator*.

4. **Data.** Pinar's bare mug cooled from 120 to 96 degrees in 30 minutes, and her foam-wrapped mug cooled from 120 to 108. How many degrees did each cool? Which one insulated better?

★ **Challenge: Explain.** A vacuum bottle keeps soup hot for hours. How does it slow conduction, convection and radiation?

Looking Ahead: Next week: People Protecting Earth. You will use real data to see how people change land, water, air and living things, and meet the scientists and communities fixing the damage.

KCE PATHWAYS · GRADES 4–5 · UNIT 6: VOICES OF CHANGE · WEEK 28 · SOCIAL STUDIES

The Cold War and the Space Race (1945–1991)

Name: _____ **Date:** _____

PRE-LESSON WARM-UP · DO THIS BEFORE YOUR COMPUTER LESSON

Pre-Lesson Warm-Up: What Do I Already Know?

Write what you know right now. Guesses are fine; this is your starting point.

1. **Review (Week 27):** Name one group that organized for equal rights in the 1960s or 1970s, and one law or win it helped bring about.

2. A "cold war" is a rivalry without a full war between the two main sides. Why might two powerful nations compete without fighting each other directly?

3. Prediction: If a rival country launched the first satellite into space, how might Americans feel, and what might they do next?

4. What do you already know about space travel or the Moon landing? Write two facts or guesses.

5. Write one question you have about the Cold War, the Space Race or the Berlin Wall.

KCE PATHWAYS · GRADES 4–5 · UNIT 6: VOICES OF CHANGE · WEEK 28 · SOCIAL STUDIES

The Cold War and the Space Race (1945–1991)

Name: _____ **Date:** _____

POST-LESSON CHALLENGE · DO THIS AFTER YOUR COMPUTER LESSON

Post-Lesson Activity: A Cold War Without a Battle

Use evidence from today's lesson. Write in complete sentences.

1. **Put in order with years:** Berlin Wall opens; Sputnik 1 launches; Berlin Airlift begins; Apollo 11 lands on the Moon; Cuban Missile Crisis.

2. **Find the mistake.** "Alan Shepard was the first person to orbit Earth, in 1957." Rewrite the sentence so it is true.

3. **Compare two sources.** The Space Act (1958) says space should be used "for peaceful purposes." Kennedy (1961) set a goal of reaching the Moon "before this decade is out." How is a law different from a speech?

4. **Write a paragraph.** Choose Katherine Johnson, Mary Jackson, Yuri Gagarin, Sally Ride or Guion Bluford. In 3 or 4 sentences, tell what barrier this person faced or broke and why it mattered. Check capitals, commas and end punctuation.

★ **Challenge: Two steps:** The Berlin Wall went up in 1961 and opened 28 years later. Germany was reunified the next year. In what year was Germany reunified? Show your work.

Looking Ahead: Next week: a changing nation (1965–today), from the Immigration Act of 1965 and the computer age to the first Earth Day and ways young people take part in civic life.